

QUESNEL, BC, Canada

WMO: 717790

Lat: 53.0269N

Lon: 122.5064W

Elev: 1789

StdP: 13.77

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.6	-9.8	-25.5	1.4	-17.2	-17.2	2.3	-8.6	19.2	23.8	16.9	23.7	0.9	0	0.591

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.4	85.7	62.3	81.7	61.0	78.1	59.7	64.5	80.2	62.6	77.4	61.0	74.7	5.0	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.0	79.7	67.5	57.3	74.9	65.7	55.8	70.8	64.1	30.5	80.2	29.0	77.4	27.9	75.2	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	14.4	12.4	-22.7	91.9	8.3	3.5	-28.7	94.4	-33.5	96.5	-38.2	98.5	-44.3	101.0
			-23.0	68.4	8.2	2.0	-29.0	69.8	-33.8	70.9	-38.4	72.0	-44.3	73.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.9	20.6	24.1	33.5	42.8	51.8	58.2	62.5	61.3	52.7	41.9	30.6	21.1	
	DBStd	17.47	14.48	12.39	10.03	6.40	6.43	5.06	5.01	5.27	6.18	6.87	10.27	12.79	
	HDD50	4218	912	725	511	231	55	2	0	0	41	264	582	895	
	HDD65	8511	1377	1145	976	666	409	214	110	138	369	716	1031	1360	
	CDD50	1250	0	0	0	15	113	246	388	352	123	12	1	0	
	CDD65	67	0	0	0	0	1	8	33	24	1	0	0	0	
	CDH74	1797	0	0	0	6	99	279	734	584	95	0	0	0	
	CDH80	518	0	0	0	0	15	70	251	163	19	0	0	0	
Wind	WSAvg	3.9	4.0	3.9	4.6	4.6	4.2	3.6	3.3	3.0	3.1	3.9	4.6	4.3	
Precipitation	PrecAvg	21.9	2.0	1.0	1.0	1.2	1.6	2.6	2.6	1.9	1.9	2.2	2.1	1.8	
	PrecMax	27.9	3.7	1.9	1.9	2.0	3.0	5.4	4.4	3.2	4.1	3.5	3.7	4.2	
	PrecMin	17.1	1.0	0.1	0.3	0.4	0.7	1.2	0.8	0.7	0.5	0.8	0.9	0.3	
	PrecStd	2.9	0.8	0.5	0.4	0.4	0.7	1.1	1.0	0.7	0.9	0.8	0.8	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.1	49.0	59.7	73.5	81.8	87.2	92.3	88.7	82.7	66.8	56.9	46.6	
		MCWB	39.8	39.7	45.9	53.1	58.0	62.0	64.2	63.1	60.2	53.9	46.6	38.9	
	2%	DB	43.5	45.2	54.7	66.2	76.5	80.9	86.5	84.5	76.3	61.2	50.9	42.6	
		MCWB	37.4	37.8	42.7	49.7	55.9	59.4	62.8	62.4	59.1	50.4	43.5	36.6	
	5%	DB	40.7	42.0	51.0	61.1	72.1	76.7	82.0	80.7	70.5	56.5	46.3	39.3	
		MCWB	35.8	35.7	40.7	46.5	54.5	58.0	61.6	61.2	57.0	48.0	40.8	35.0	
	10%	DB	37.1	38.9	47.5	56.8	67.6	72.4	77.6	76.5	65.4	53.1	43.0	36.4	
		MCWB	33.9	34.0	38.9	44.1	52.6	56.7	60.3	60.0	54.2	46.0	38.5	33.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.6	41.1	47.2	53.6	60.2	64.4	67.1	67.1	62.3	55.5	47.6	40.3
			MCDB	45.9	47.2	58.0	70.2	77.3	81.5	85.8	82.0	76.8	64.1	54.3	44.8
2%		WB	37.9	38.5	43.6	50.6	57.6	61.5	64.7	64.6	60.0	52.0	44.5	37.4	
		MCDB	42.4	43.8	52.6	63.9	72.4	76.4	81.5	78.9	73.0	58.6	49.6	41.7	
5%		WB	36.0	36.6	41.4	47.8	55.6	59.4	62.8	62.5	57.9	49.1	41.6	35.3	
		MCDB	40.0	41.0	49.7	58.9	68.9	72.5	77.6	77.2	68.6	54.9	45.6	38.5	
10%		WB	34.1	34.6	39.3	45.3	53.8	57.7	61.2	60.7	55.7	46.6	39.0	33.3	
		MCDB	36.7	38.2	46.6	54.9	65.6	69.5	74.7	74.0	63.8	52.1	42.3	36.1	
Mean Daily Temperature Range	5% DB	MDBR	13.2	16.9	20.1	23.9	25.6	24.2	25.4	26.7	23.6	18.0	11.8	12.2	
		MCDBR	14.2	18.2	24.5	32.1	35.4	34.1	36.2	35.6	33.2	24.0	16.2	13.5	
		MCWBR	10.1	12.4	14.6	17.6	17.3	15.3	15.2	15.8	17.7	14.8	10.8	9.9	
	5% WB	MCDBR	13.4	16.4	23.3	28.4	30.7	29.6	31.6	32.5	30.2	21.7	14.8	12.4	
		MCWBR	10.1	11.7	14.4	16.0	15.6	13.9	14.2	15.2	16.6	14.0	10.4	9.6	
Clear-Sky Solar Irradiance	taub	0.269	0.277	0.298	0.328	0.350	0.358	0.348	0.353	0.318	0.300	0.285	0.266		
	taud	2.361	2.402	2.452	2.432	2.419	2.413	2.486	2.459	2.564	2.582	2.471	2.330		
	Ebn at Noon	232	269	285	288	285	283	284	277	276	259	226	208		
	Edh at Noon	18	23	28	32	34	35	32	31	25	20	16	15		
All-Sky Solar Radiation	RadAvg	271	578	1033	1494	1720	1872	1871	1596	1084	624	292	204		
	RadStd	25	60	70	98	119	140	176	69	93	52	30	20		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (6 neighbors)	N/A	N/A	N/A	N/A	-0.50	N/A	N/A	N/A	+137	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page